

Co-existence of plasmid-mediated *bla*_{NDM-1} and *bla*_{NDM-5} in *Escherichia coli* sequence type 167 and ST101 and their discrimination through restriction digestion

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ABSTRACT The concurrent presence of multiple New Delhi metallo-β-lactamase (*bla*_{NDM}) variants within an isolate often goes undetected without next-generation sequencing. This study detects and characterizes dual *bla*_{NDM} variants in *Escherichia coli* through Sanger and whole-genome sequencing. Additionally, a rapid identification method utilizing restriction digestion was designed for detecting *bla*_{NDM} variants carrying M154L mutation. Antibiotic susceptibility, minimal inhibitory concentration for meropenem and ertapenem, PCR, and Sanger sequencing of *bla*_{NDM} along with genome sequencing using Illumina and Nanopore technology were conducted. Transmissibility and replicon types of *bla*_{NDM}-harboring plasmids were evaluated. Restriction digestion using restriction enzyme, BtsCI was developed to distinguish between *bla*_{NDM-1} and *bla*_{NDM} variants possessing M154L mutation, such as *bla*_{NDM-5}, *bla*_{NDM-7} etc. Two isolates belonging to phylogroups A; ST167 and B1; ST101 and resistant to meropenem and ertapenem (≥16 mg/L) were recovered from the blood of a neonate and the rectal swab of a pregnant woman, respectively. *bla*_{NDM} was detected by PCR, and Sanger sequences of *bla*_{NDM} showed two peaks at 262 (G and T) and 460 (A and C) nucleotide positions indicative of more than one *bla*_{NDM} variant. Hybrid assembly confirmed co-existence of *bla*_{NDM-1} and *bla*_{NDM-5} in each isolate. *bla*_{NDM-1} was located on IncY (ST167) and IncHI1A/HI1B (ST101), while *bla*_{NDM-5} was on IncFIA/FII (ST167) and IncC (ST101) plasmids in the two isolates. Digestion with BtsCI could discriminate between *bla*_{NDM-1} and *bla*_{NDM-5}. The co-existence of multiple *bla*_{NDMs}, *bla*_{NDM-1}, and *bla*_{NDM-5} in epidemic clones of *E. coli* is concerning. Restriction digestion method and Sanger sequencing can facilitate quick identification of dual *bla*_{NDM} variants in a single isolate.

IMPORTANCE The global dissemination of antimicrobial resistance genes is a serious concern. One such gene, *bla*_{NDM}, has spread globally via plasmids. *bla*_{NDM} confers resistance against all β-lactam antibiotics, except monobactams. Most of the earlier literature reported the presence of single *bla*_{NDM} variant. However, this study reports the prevalence of dual *bla*_{NDM} variants (*bla*_{NDM-1} and *bla*_{NDM-5}) located on two separate plasmids identified in two distinct *Escherichia coli* epidemic clones ST167 and ST101 isolated from a septicemic neonate and a pregnant mother, respectively. *bla*_{NDM-5} differs from *bla*_{NDM-1} due to the presence of two point mutations (i.e., V88L and M154L). This study detected dual *bla*_{NDM} variants through Sanger sequences and further validated them through hybrid-genome assembly. Detection of multiple *bla*_{NDM} variants in a single isolate remains difficult until genome sequencing or southern blotting is carried out. Hence, a simple restriction digestion method was devised to rapidly screen dual *bla*_{NDM} variants containing M154L mutation.

KEYWORDS neonatal sepsis, epidemic clone, dual *bla*_{NDM} variants, *bla*_{NDM-1,-5}, nanopore, India

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New Delhi metallo- β -lactamase (bla_{NDM}) is the fastest and most widespread carbapenemase that has triggered an alarming threat worldwide since its identification in 2009 (1, 2). Around 71 variants of bla_{NDM} have been reported worldwide, of which bla_{NDM-1} , bla_{NDM-5} , and bla_{NDM-7} are prevalent (3). NDM confers resistance against all β -lactam antibiotics, including carbapenems, the drug of last resort (4, 5).

Most studies report the presence of a single bla_{NDM} variant in a bacterial genome with few exceptions where two copies of bla_{NDM} -variants ($bla_{NDM-1/NDM-5}$) were present either in chromosome or plasmids in a single bacteria, such as *Pseudomonas aeruginosa*, *Escherichia coli*, *Klebsiella michiganensis*, and *Acinetobacter johnsonii* (Table S1) (6–12). *E. coli*, being a commensal and an opportunistic pathogen, acts as a reservoir of acquired antimicrobial resistance determinants eventually transferring it to other species (13).

In this study, we identified two *E. coli* isolates from blood (neonate) and rectal swab (adult) possessing two different variants of bla_{NDM} . To the best of our knowledge, this is the first report where carriage of bi-variant bla_{NDM} (bla_{NDM-1} and bla_{NDM-5}) in *E. coli* is being reported along with their characterization and genome analysis.

Two clinical *E. coli* isolates EN5349 and IN-MR210EC, were recovered from the blood of a septicemic neonate and the rectal swab of a hospitalized pregnant mother [as part of a collaborative study, called Burden of antibiotic resistance in neonates from developing societies (BARNARDS)], respectively. Isolates were assessed for antibiotic susceptibility by disk diffusion assay, and minimal inhibitory concentrations (MICs) for meropenem, ertapenem, and colistin (Sigma-Aldrich, Steinheim, Germany) were determined by the broth-micro dilution (14). PCR amplicons of bla_{NDM} were sequenced using primer pairs (Table S2) in an Applied Biosystems DNA analyzer (Perkin Elmer, USA) (15). Both short (Illumina NextSeq 500 platform, San Diego, CA) and long read-based (Oxford Nanopore, UK) sequencing technologies were used for genome sequencing. Unicycler was used to generate hybrid assemblies, which were further used for downstream analysis (Supplementary methods) (16). From hybrid assemblies, bla_{NDM} harboring plasmids were constructed and searched in the bacterial plasmid database (PLSDB) for similar complete plasmid sequences (17). Plasmids showing nucleotide identity ($\geq 99\%$) and same replicon types with similar bla_{NDM} variants with reference to study plasmids were compared.

Transfer of bla_{NDM} into *E. coli* J53 Az^r strain (100 mg/L) was attempted by solid-mating conjugation assay and electro-transformation into *E. coli* DH10B cells (Invitrogen, CA, USA) (Supplementary methods) (14, 15). Plasmid replicon types were determined by PCR-based replicon typing (PBRT, Diatheva, Italy) for wild type isolates (WTs) and transformants (TFs) (18, 19). PCR amplicons of bla_{NDM} gene were digested for 4 h using the BtsCI enzyme (New England Biolabs, USA) at 50°C.

Isolates belonged to phylogroups A; ST167 (EN5349) and B1; ST101 (IN-MR210EC), which are epidemic clones conferring resistance to carbapenems and other antibiotics while remaining susceptible to tigecycline and colistin (20). Presence of bla_{NDM} in isolates was confirmed by Sanger sequencing. Chromatogram of sequences (forward and reverse) for both isolates depicted the presence of two sharp peaks at 262 (G and T) and 460 (A and C) positions. Two different base calls at a single position suggested amplification of more than one bla_{NDM} in a single isolate (Fig. S1). G at position 262 and A at 460 corresponded to sequence of bla_{NDM-1} , whereas G262→T (V88L) and A460→C (M154L) matched with bla_{NDM-5} (21). Meticulous evaluation of chromatogram helped the detection of two distinct bla_{NDM} copies.

Genome sequencing confirmed that isolates carried two copies of bla_{NDM} (bla_{NDM-1} and bla_{NDM-5}), along with two copies of bla_{TEM-1B} , which may increase enzyme production and enhance the chance of spread through various plasmids (15, 22). Both isolates EN5349 and IN-MR210EC exhibited a complex resistance profile, harboring multiple resistance determinants. These included genes conferring resistance to β -lactams (bla_{NDM-1} , bla_{NDM-5} , bla_{TEM-1B} , $bla_{CTX-M-15}$, bla_{CMY-42} , bla_{OXA-2} , bla_{OXA-9} , bla_{CMY-6}), aminoglycosides [*rmtB*, *rmtF*, *armA*, *aadA1*, *aadA2*, *aac(6')-Ib*, *aac(6')-Ib-cr*, *aph(6)-Id*, *aph(3'')-Ib*, *aac(3)-IIa*], sulfonamides (*sul1*, *sul2*), trimethoprim (*dfrA12*, *dfrA29*), phenicols (*catA1*), fluoroquinolones (*qnrS1*), and efflux pumps [*qacE*, *msr(E)*, *mph(E)*]

TABLE 1 Genome-based characterization of EN5349 and IN-MR210EC along with characterization of transformants (TFs) in terms of resistance determinants and plasmids^a

Isolate/Accession number	EN5349/JAPTGK000000000	IN-MR210/JAYKKU000000000
Date of isolation	May, 2017	August, 2017
Source	Blood	Rectal swab
Phylogroup/Sequence types	A/ST167	B1/ST101
Virulence determinants		
Adherence	<i>ecpABCDER, elfACDGG, eaeH, etpA, hcpABC, fimDFG, pilW</i>	<i>cfaABCDE, ecpABCDER, elfACDGG, eaeH, hcpABC, fimABCDEF, fliG, fliC, mrkABD</i>
Autotransporter	<i>cah, ehaB</i>	<i>ehaAB, upaG/ehaG</i>
Invasion	<i>ibeBC, tia</i>	<i>ibeBC, tia</i>
Iron uptake	<i>sirAD, fyuA, irp1, irp2, ybtAE PQSTUX, hemB</i>	<i>fyuA, irp1, irp2, ybtAE PQSTUX</i>
Toxin	<i>hlyE/clyA</i>	<i>hlyE/clyA</i>
Serum resistance	<i>iss, traT</i>	<i>iss</i>
Others	<i>Wzi, stjC, gale, mntB</i>	<i>lptBCE</i>
Serotype/C-H type	O101:H9/11-0	O131:H31/41-191
Antibiotic susceptibility	Piperacillin ^R , Cefotaxime ^R , Ciprofloxacin ^R , Trimethoprim-sulfamethoxazole ^R , Aztreonam ^R , Amikacin ^R , Gentamicin ^R , Meropenem ^R , Colistin ^S , Tigecycline ^S	Piperacillin ^R , Cefotaxime ^R , Ciprofloxacin ^R , Trimethoprim-sulfamethoxazole ^R , Aztreonam ^R , Amikacin ^R , Gentamicin ^R , Meropenem ^R , Colistin ^S , Tigecycline ^S
Minimum inhibitory concentration (MIC)		
Meropenem	128 mg/L	16 mg/L
Ertapenem	64 mg/L	64 mg/L
Size of genome	5253 kb	5417 kb
	Chromosome 5014 kb	Chromosome 4820 kb
	Extra-chromosomal element 315 kb	Extra-chromosomal element 601 kb
GC content (%)	50.80%	50.40%
Core genome sequence type (cgST)	cgST169598	cgST28992
Resistance determinants (detected through PCR and WGS)		
Wild type	<i>bla_{NDM-1}, bla_{NDM-5}, bla_{TEM-1B}, bla_{CTX-M-15}, bla_{CMY-42}, rmtB, rmtF, aadA2, sul1, sul2, dfrA12, ARR-2, qacE, tet(A), mph(A), qnrS1, aac(6)-Ib-cr, aph(6)-Id, aph(3'')-Ib</i>	<i>bla_{NDM-1}, bla_{NDM-5}, bla_{TEM-1A}, bla_{TEM-1B}, bla_{TEM-1B}, bla_{OXA-2}, bla_{OXA-9}, bla_{CMY-6}, armA, rmtB, aadA1, aadA2, sul1, sul1, dfrA12, dfrA29, qacE, qacE, qacE, msr(E), mph(E), catA1, aac(6)-Ib, aac(6)-Ib-cr, aph(6)-Id, aph(3'')-Ib, aac(3)-IIa</i>
Resistance determinants (detected through PCR)		
TF1 (<i>bla_{NDM-1}</i> ^{+ve})	<i>bla_{NDM-1}, bla_{TEM-1}, bla_{CTX-M}, bla_{CMY}, qnrS, aac(6)-Ib-cr</i>	<i>bla_{NDM-1}, bla_{TEM-1}, bla_{TEM-1B}, aac(6)-Ib</i>
TF2 (<i>bla_{NDM-5}</i> ^{+ve})	<i>bla_{NDM-5}, bla_{TEM-1}, bla_{CMY}, rmtB</i>	<i>bla_{TEM-1}, rmtB</i>
TF3 (<i>bla_{NDM-1}</i> ^{+ve} , <i>bla_{NDM-5}</i> ^{+ve})	<i>bla_{NDM-1}, bla_{NDM-5}, bla_{TEM-1}, bla_{CTX-M}, bla_{CMY}, rmtB, qnrS, aac(6)-Ib-cr</i>	NF
Plasmid replicon types (PBRT and WGS)		
Wild type	ColRNAI, IncFIA, IncFII, IncY, IncY	IncC, IncHI1A, IncHI1B
TF1 (<i>bla_{NDM-1}</i> ^{+ve})	IncY	IncC
TF2 (<i>bla_{NDM-5}</i> ^{+ve})	IncFII	IncHI1A, IncHI1B

(Continued on next page)

TABLE 1 Genome-based characterization of EN5349 and IN-MR210EC along with characterization of transformants (TFs) in terms of resistance determinants and plasmids^a (Continued)

Isolate/Accession number	EN5349/JAPTGK000000000	IN-MR210/JAYKKU000000000
TF3 (<i>bla</i> _{NDM-1} ⁺ , <i>bla</i> _{NDM-5} ⁺)	IncFII, IncY	NF
MIC of Meropenem		
TF1 (<i>bla</i> _{NDM-1} ⁺)	64 mg/L	8 mg/L
TF2 (<i>bla</i> _{NDM-5} ⁺)	16 mg/L	8 mg/L
TF3 (<i>bla</i> _{NDM-1} ⁺ , <i>bla</i> _{NDM-5} ⁺)	32 mg/L	NF
MIC of Ertapenem		
TF1 (<i>bla</i> _{NDM-1} ⁺)	64 mg/L	32 mg/L
TF2 (<i>bla</i> _{NDM-5} ⁺)	64 mg/L	32 mg/L
TF3 (<i>bla</i> _{NDM-1} ⁺ , <i>bla</i> _{NDM-5} ⁺)	32 mg/L	NF
gyrA-parC mutations	<i>gyrA</i> :p.S83L, <i>gyrA</i> :p.D87N, <i>parC</i> :p.S80I, <i>parE</i> :p.S458A	<i>gyrA</i> :p.S83L, <i>gyrA</i> :p.D87N, <i>parC</i> :p.S80I, <i>parE</i> :p.E460D
Integrons	In27, In406	In27, In573
Characterization of <i>bla</i> _{NDM-1} -carrying plasmid		
Plasmid Id	P1-EN5349	P1-IN-MR210EC
Size	122 kb	265 kb
Addition system	relBE	relBE
Replicon type	IncY	IncHI1A, IncHI1B
Carriage of additional resistance determinants	<i>bla</i> _{TEM-1B} , <i>bla</i> _{CTXM-15} , <i>rmtF</i> , <i>sul2</i> , <i>ARR-2</i> , <i>qnrS1</i> , <i>aac</i> -(6')-Ib-cr, <i>aph</i> -(6)-Id, <i>aph</i> -(3'')Ib	<i>bla</i> _{TEM-1A} , <i>bla</i> _{OXA-9} , <i>Δbla</i> _{DHA-1} , <i>armA</i> , <i>aadA</i> , <i>sul1</i> , <i>qacE</i> , <i>msr</i> (E), <i>mph</i> (E), <i>aac</i> -(6')-Ib, <i>aac</i> -(6')-Ib-cr
Genetic Environment	IS3000 → <i>ΔISAba125</i> → <i>bla</i> _{NDM-1} → <i>ble</i> _{MBL} → <i>trpF</i> → <i>dsbD</i> → <i>groE5</i> → <i>groEL</i> → IS3000	IS26 → <i>ΔISAba125</i> → <i>bla</i> _{NDM-1} → <i>ble</i> _{MBL} → <i>Δbla</i> _{DHA-1} → <i>ampR</i> → <i>sul1</i> → ISCR1 → IS5
Characterization of <i>bla</i> _{NDM-5} -carrying plasmid		
Plasmid Id	P2-EN5349	P2-IN-MR210EC
Size	125 kb	137 kb
Addition system	<i>penKI</i>	NF
Replicon type	FIA/FII (F36:A4:B-)	IncC
Carriage of additional resistance determinants	<i>bla</i> _{TEM-1B} , <i>rmtB</i> , <i>aadA2</i> , <i>sul1</i> , <i>dfrA12</i> , <i>qacE</i> , <i>tetA</i> , <i>mph</i> (A)	<i>bla</i> _{TEM-1B} , <i>bla</i> _{CMY-6} , <i>rmtB</i> , <i>aadA2</i> , <i>sul1</i> , <i>dfrA12</i> , <i>qacE</i>
Genetic Environment	IS26 → <i>ΔISAba125</i> → <i>bla</i> _{NDM-5} → <i>ble</i> _{MBL} → <i>trpF</i> → <i>dsbD</i> → ISCR → <i>sul1</i> → <i>aadA</i> → DuF1010 → <i>dfrA12</i> → <i>IntI</i> → IS26	IS26 → <i>ΔISAba125</i> → <i>bla</i> _{NDM-5} → <i>ble</i> _{MBL} → <i>trpF</i> → <i>dsbD</i> → ISCR1 → <i>sul1</i> → <i>aadA2</i> → DuF1010 → <i>dfrA12</i> → <i>IntI1</i> → TnAs2

^aR, resistant; S, susceptible; TF, transformant; Inc, incompatibility; IS, insertion sequence; Int, integrase; NF, not found.

(Table 1). Additionally, IN-MR210EC carried multiple copies of *bla*_{TEM-1A}, *sul1*, and *qacE* (Table 1). Presence of multiple resistance determinants indicated that several antibiotics would be ineffective against these isolates, implying a potential risk of treatment failure with the targeted antibiotics. The isolates exhibited susceptibility to tigecycline and colistin. However, the clinical utility of these two antibiotics is often limited due to potential adverse effects, such as nephrotoxicity with colistin and gastrointestinal disturbances with tigecycline (23, 24).

Hybrid genome assembly revealed that *bla*_{NDM-1} and *bla*_{NDM-5} were harbored in two distinct plasmids, viz. *bla*_{NDM-1} in IncY [P1-EN5349; 122 kb] and IncHI1A/HI1B [P1-IN-MR210EC; 265 kb], whereas *bla*_{NDM-5} were in IncFIA/FII [P2-EN5349; 125 kb] and IncC [P2-IN-MR210EC; 137 kb] (Fig. 1). Earlier studies have reported the occurrence of double copies of same *bla*_{NDM} variant, either located in the chromosome or in plasmids, among different organisms (Table S1) (6–12). This study reports different *bla*_{NDM} variants in a single *E. coli* isolate.

The genetic environment of *bla*_{NDM-1} featured a truncated *ISAba125* upstream, along with Tn3-like *IS3000* and *IS26*-like family transposase located further upstream of *ISAba125* in EN5349 and IN-MR210EC, respectively. The downstream regions of *bla*_{NDM-1} varied (Table 1). Moreover, for *bla*_{NDM-5}, both isolates possessed an *IS26*-like family transposase and Δ *ISAba125* upstream, with *ble*_{MBL} downstream, followed by *trpF*, *dsbD*, and *IS91*-like *ISCR1* family transposase (Table 1). The genetic environment of *bla*_{NDM-5} in study plasmids, P2-EN5349 and P2-IN-MR210EC showed >95% similarity (Fig. S2). Genetic environments of the study *bla*_{NDM} variants were individually comparable with the genetic backgrounds of the respective *bla*_{NDM} variants reported worldwide (4, 15, 25).

Conjugation of study plasmids was unsuccessful, and transformants obtained by electroporation carried either *bla*_{NDM-1} or *bla*_{NDM-5}. Few TFs of EN5349 (EN5349.TF3) co-harbored *bla*_{NDM-1} and *bla*_{NDM-5} as confirmed by Sanger sequencing. WT and TFs exhibited high MIC values for meropenem and ertapenem (≥ 16 mg/L) (Table 1). *bla*_{NDM} has been known to be promiscuous in nature, and it is easily transmitted between organisms (4, 26). Study plasmids, such as P2-EN5349, P1-IN-MR210EC, and P2-IN-MR210EC, possessed plasmid transfer/mobilization factors (*tra*-operon system, *mobI*), and P1-EN5349 possessed conjugation/type IV secretion system (T4SS) (*virB*, *virB9*, *virB11*, *virB6*, *virB4*, *Rhs*) (Fig. 1) (27, 28) analyzed through genome sequences. Although conjugation was unsuccessful under laboratory conditions, the presence of such genes in plasmids still suggests the potential for *bla*_{NDM} transmission in natural environments. Such transmission may occur not only in the hospital environment but also in other environments. Multiple *bla*_{NDM} variants in *E. coli* epidemic clones increase the potential for transmission, along with transfer of other resistance genes present in the same plasmid (14, 20).

Study plasmids showed close resemblance with some globally reported plasmids harboring similar *bla*_{NDM} variants and same replicon types in different species of Enterobacterales. P2-EN5349 (pMLST F36:A4:B-) showed similarity with 11 globally reported IncFIA/FII plasmids harboring the *bla*_{NDM-5} gene. These plasmids were identified in diverse *E. coli* isolates obtained from clinical specimens (rectal swabs, urine, faecal swabs, and mastitis milk) across various countries, including Canada, the Czech Republic, Switzerland, Italy, Myanmar, Bangladesh, China, Thailand, and India (Fig. 2; Table S3) (17). In contrast, P2-IN-MR210EC (*bla*_{NDM-5} in IncC plasmid) exhibited similarity only to a plasmid from environmental *E. coli* found in Switzerland. These findings suggest a more widespread dissemination of *bla*_{NDM-5} via IncFIA/FII plasmids compared to IncC. There were very few reports of *bla*_{NDM-1} transmission via IncY and IncHI1A/HI1B plasmids. Notably, P1-IN-MR210EC (*bla*_{NDM-1} in IncHI1A/HI1B) shared similarities with five plasmids found in clinical *K. pneumoniae* isolates from India, Thailand, and New Zealand. (Fig. 2; Table S3) (17).

With the emergence of organisms carrying two different variants of *bla*_{NDM} in different plasmids, spread of such variants and their stability in antibiotic-loaded

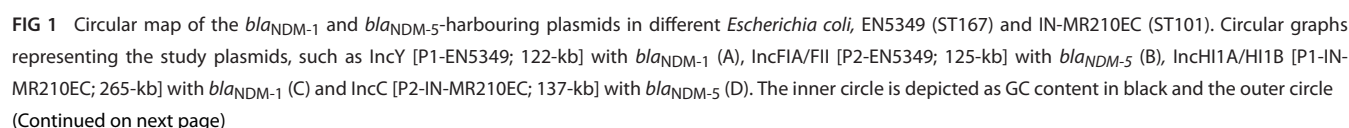


Fig 1 (Continued)

represents GC skew in grey and purple. Resistance gene is denoted in red, coding sequences (CDS) in blue, prophage regions (phigaro) in brown, genes involved in transfer is in orange and integration/excision is in green.

environment calls for timely identification of such strains. Dual *bla*_{NDM} variants in a single isolate may be confirmed by southern blotting and hybrid assembly, which are expensive, time-consuming, and difficult to execute for a diagnostic laboratory. Hence, a novel restriction digestion-based method was introduced to distinguish between *bla*_{NDM} variants (NDM-4, 5, 7, 8, 12, 13, 15, 16b, 17, 19, 20, 21, 27, 35, 36, 37) with or without M154L mutation. *bla*_{NDM-1} has two recognition sites for BtsCI (Fig. S3), which generate three DNA fragments (255, 215, and 355 bp). Presence of a mutation (A460→C) corresponding to M154L alters the second recognition site (GGATG→GGCTG), which BtsCI is unable to cleave, resulting in two fragments of 255 and 570 bp (Fig. S3). Hence, most M154L possessing variants (including *bla*_{NDM-5}) will produce fragments different from the *bla*_{NDM-1} variant. Since study isolates possessed both *bla*_{NDM-1} and *bla*_{NDM-5}, four DNA fragments, that is, 215, 255, 355, and 570 bp (Fig. S3), were generated, which confirmed the presence of *bla*_{NDM-1} along with a variant possessing M154L mutation (4). Identification of resistance genes in diagnostic laboratories, especially in low- and middle-income countries (LMICs), still remains limited due to the lack of access to PCR-based molecular techniques. Diagnostic laboratories rely on conventional disk diffusion tests or MIC values generated through automated systems, which further guide their treatment decisions. While whole-genome sequencing (WGS) offers a comprehensive approach for identifying resistance determinants, its high cost and lack of expertise in data analysis limit its widespread adoption in LMIC settings.



FIG 2 Worldwide prevalence of plasmids similar to study plasmids. Globally reported plasmids similar ($\geq 99\%$ nucleotide identity) to study plasmids are denoted in four different colours such as yellow (IncY with *bla*_{NDM-1}), red (IncFIA/FII with *bla*_{NDM-5}), green (IncHI1A/HI1B with *bla*_{NDM-1}) and sky blue (IncC with *bla*_{NDM-5}). The map was obtained from freeworldmaps.net.

Our study adds to the very few studies that reported more than one variant of *bla*_{NDM} in a single isolate. The presence of multiple *bla*_{NDM} variants in *E. coli* epidemic clones collected from a septicemic neonate and a pregnant mother (not paired) is worrisome. Emergence and spread of such organisms are of immense public health consequence. Two different variants of *bla*_{NDM} in different plasmids and their stability in antibiotic-loaded environment calls for timely identification.

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S.B. contributed to the conception of the work, analysis, interpretation of data, and drafting of the manuscript. A.B. performed all experimental work and contributed to the acquisition of laboratory and W.G.S. data with manuscript drafting. P.B. and J.S. performed the W.G.S. data analysis with manuscript editing. S.M. contributed to the analysis of data. S.D. contributed to the review of data. The final manuscript was read and approved by all authors.

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DATA AVAILABILITY

Genome data were submitted to the NCBI database with accession numbers [JAPTGK000000000](#) and [JAYKKU000000000](#) (Table 1).

ETHICAL APPROVAL

This study was conducted on archived strains; hence, ethical approval was not required.

ADDITIONAL FILES

The following material is available [online](#).

Supplemental Material

Supplemental material (Spectrum00987-24-s0001.docx). Supplemental methods, Tables S1 to S3, and Figs. S1 to S3.

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